(Please write your Enrollment Number) Enrollment No.

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:< B. Tech. (CSE /IT /CSEAI / AI&ML) >, Semester:<4th>
(May, 2024)

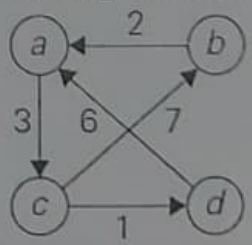
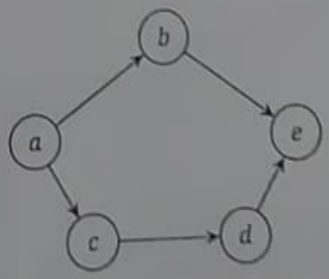
Subject Code: BCS 204	Subject: Design and Analysis of Algorithms
Time :3 Hours	Maximum Marks :60
Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.	

Q1			(2.5*8=20)	
	(a)	Solve the recurrence, $T(N) = 25T(N/5) + N^{2/2}$		
	(b)	Given N distinct integers, prove the lower bound of $\left\lceil \frac{3N}{2} - 2 \right\rceil$ comparisons in the worst case to determine both the least and greatest element.		
	(c)	Does Prim's and Kruskal's algorithm always work correctly on graphs with negative edge weights? Justify your answer.		
	(d)	Compare the Divide and Conquer strategy with Dynamic Programming strategy.		
	(e)	If any sparse graph is a graph in which $\|E\| \in O(\|V\|)$, which implementation of DFS will have a better time efficiency for such graph, the one that uses the adjacency matrix or the one that uses the adjacency lists? Justify your answer in terms of associated complexities.		
	(f)	Generate a state space tree using backtracking that provides the all permutations of $\{1, 2, 3, 4\}$.		
	(g)	In case of string matching, if all characters in the pattern 'P' are different then suggest a method to accelerate Naïve-String-Matching to run in time $O(N)$ on an N-character text 'T'.		
	(h)	Suppose that two problems are known to be NP-complete. Does this imply that there is a polynomial time reduction from one to the other? Justify.		
		UNIT-I		
Q2	(a)	Solve the recurrence : $T(N) = 4T(\sqrt{N}) + \lg^{3}_{6} N$	(5*2=10)	
	(b)	Apply Strassen's algorithm to compute the following matrix multiplication: $\begin{bmatrix} 1 & 0 & 2 & 1 \\ 4 & 1 & 1 & 0 \\ 0 & 1 & 3 & 0 \\ 5 & 0 & 2 & 1 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 0 & 1 \\ 2 & 1 & 0 & 4 \\ 2 & 0 & 1 & 1 \\ 1 & 3 & 5 & 0 \end{bmatrix}$ Clearly mention each step of the computation.		
Q3	(a)	Consider a grid of 1 x N squares and assume that each square is colored white or black such that no two adjacent squares are colored white. Let T(N) be the recurrence that denotes the number of different ways that this grid can be colored. Derive the recurrence equation of this recurrence and solve it.	(5*2=10)	
	(b)	Provide divide-and-conquer recurrence relation that describes the running time T(n) of the Quick Sort algorithm and Merge Sort algorithm. Also, provide their asymptotic solution for the running time.		
		UNIT-II		
Q4	(a)	Find an optimal parenthesization of a matrix-chain product whose sequence of dimensions is (5x10; 10x3; 3x12; 12x5; 5x50; 50x6).	(5*2=10)	
	(b)	What is an optimal Huffman code for the following set of frequencies, based on the first 8 Fibonacci numbers? a:1, b:1, c:2, d:3, e:5, f:8, g:13, h:21. Can you generalize your answer to find the optimal code when the frequencies are the first n Fibonacci numbers?		

Q5	(a) Modify Kruskal algorithm as follows: sort the edges of graph in decreasing order, and then run the algorithm as before. Is this modified algorithm produces a maximum cost spanning tree? If true then prove your answer, and if false then give a counter example. (b) Determine the cost and structure of an optimal binary search tree for a set of n=7 keys with the following probabilities:	(5*2=10)	

i	0	1	2	3	4	5	6	7
p_i		0.04	0.06	0.08	0.02	0.10	0.12	0.14
q_i	0.06	0.06	0.06	0.06	0.05	0.05	0.05	0.05

UNIT-III

Q6	(a) Solve the all-pairs shortest-path problem using Floyd-Warshall algorithm for the given below directed graph:  (b) Consider the given below directed acyclic graph. How many topological orderings does it have? Prove your answer.  	(5*2=10)	
Q7	(a) What is N-Queen problem? Solve 5-queen problem using backtracking. (b) Compare backtracking technique with branch and bound technique using a suitable example.	(5*2=10)	

UNIT-IV

Q8	(a) How Knuth Morris Pratt (KMP) algorithm works? Using KMP algorithm find whether the pattern P=(0010) is in the text T=(1100011010001010) or not ? (b) Consider the Rabin-Karp string matching algorithm with a prime number q=13. The pattern to be searched is "ABC" and the text in which the pattern is to be searched is "ABABCABABCACABC". Perform the Rabin-Karp algorithm step by step and show the hash values at each step.	(5*2=10)	
Q9	Select any NP-Complete problem of your choice and accomplish followings: (a) Prove that the above selected problem is NP-Complete. (b) Briefly explain any existing solution for the above selected NP-Complete problem.	(5*2=10)	